Problem Statement

Given delivery data for a couple hours in the Bay Area in 2015, write a solver that assigns deliveries to drivers to maximize the average deliveries/hour across drivers, with a target level of service of average delivery duration being < 45 minutes. The deliverables we'd like are:

- For your solution, the average deliveries/hour across all drivers and average delivery duration across all deliveries. Note that your solution must take < 1 minute to run.

Data

- **Delivery ID**: Unique identifier for a delivery in the system
- **Created at**: The datetime when the order was submitted into the system. Note that times are given in UTC.
- **Food ready time**: The datetime at which the food was ready to be picked up. Note that times are given in UTC.
- **Region ID**: We split up regions, e.g. Palo Alto, Mountain View, etc. The region ID tells to which region a delivery belongs; you can use this if you'd like in your solution, but not required.
- **Pickup lat**: Latitude of the pickup location for the delivery (the restaurant)
- **Pickup long**: Longitude of the pickup location for the delivery (the restaurant)
- **Dropoff lat**: Latitude of the dropoff location for the delivery (the consumer's address)
- **Dropoff long**: Longitude of the dropoff location for the delivery (the consumer's address)

Details/Assumptions

- Delivery duration for a delivery = (time the delivery is dropped off) - (delivery.created_at). Average delivery duration = sum(delivery duration for each delivery) / (total # of delivs)
- Note that some deliveries will necessarily take longer than 45 minutes. This is ok, as the target level of service of < 45min applies to the *average* delivery duration.
- Average deliveries/hour = (total deliveries delivered) / sum(time spent on each route)
- A "route" is the sequence of pickups and drop-offs given to a particular driver. For example, [Pickup Delivery 1, Dropoff Delivery 1, Pickup Delivery 2, Dropoff Delivery 2] is an example of a route, and the total time for the route is (Time of last dropoff - Time when the driver started the route)
- All routes must start at 2/3/15 2:00 UTC (when the first few deliveries are created). So if a driver's first action is to pick up a delivery ready at 3:00 UTC, assume they have to wait from 2:00 UTC until 3:00 UTC.
- Just to be explicit, deliveries need to be picked up before they can be dropped off.
- Assume drivers travel in straight lines, with distance given by the haversine formula (note that that formula is for kilometers, can feel free to grab this code snippet online in your language of choice) at a rate of 4.5 meters/second
- Assume there are no capacity constraints for drivers.
- Assume that we don't have to worry about the drivers' original locations (i.e. they can start at the location of the store where they have their first pickup)
- Assume food is ready for a given delivery at delivery.food_ready_time, no earlier.
- Don't worry about the stochastic element present in the real-world problem, i.e. you can assume that you know about all deliveries from the very beginning.
- Can you achieve 2 or more deliveries per hour given the constraints?

Bonus question: Can you solve the problem using fewer than 50 drivers? Note that this is not required for submission